

# Notes: D’Acunto and Rossi (RFS, 2022)

## Regressive Mortgage Credit Redistribution in the Post-Crisis Era

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# 1 Big picture

- **Question.** Why did U.S. mortgage origination shift away from small and midsize loans toward large loans after the 2008–2009 financial crisis, and was this shift driven by **credit supply** or by changes in borrower demand?
- **Setting.** The paper studies the quasi-universe of U.S. mortgage applications from **2007–2017**, covering both traditional lenders and FinTech/nonbank originators, and focuses on the post-crisis period from **2011 onward**.
- **Core challenge.** Raw mortgage-origination patterns mix together borrower demand, lender supply, local macro conditions, and incentives created by conforming loan limits. The paper needs to show that the shift toward larger loans is not just a demand composition story.
- **Main contribution.** The paper combines:
  1. large-scale descriptive evidence on the full loan-size distribution;
  2. approval-rate regressions that use **rejected applications** to hold fixed observable mortgage demand;
  3. quantile regressions to track how the entire distribution of approved loan sizes moves over time; and
  4. an **instrumental-variable strategy** based on pre-crisis county exposure to large lenders.
- **Four headline facts.**
  1. Since 2011, the number, size, and approval rate of **small and medium-sized loans** have fallen relative to large loans.
  2. **Top lenders** shifted their origination behavior the most.
  3. The shift is **monotonic in lender size**; it is not driven by one arbitrary size threshold.
  4. The effects are strongest for loans **far from the conforming loan limit(s)** on both sides, so the results are not only about bunching right around the conforming cutoff.
- **Why it matters.** Mortgage-market changes affect who can buy homes, where credit flows geographically, household wealth accumulation, and potentially entrepreneurship and local growth. A systematic move away from small loans is therefore distributionally important.
- **Main empirical question.** Did larger lenders become systematically more willing to approve and originate larger mortgages after 2010, even after holding fixed borrower observables and local demand conditions?
- **Main empirical results.**
  - The post-2010 mortgage market experienced a **regressive redistribution of credit**: lending shifted from smaller toward larger loans.
  - In approval regressions, larger loans become significantly more likely to be approved after 2010, and the effect is **stronger for top-five lenders**.
  - Quantile regressions show that top lenders shift the entire approved-loan distribution to the right, especially in the **upper tail**; smaller lenders move mostly within the conforming segment.

- Instrumental-variable estimates based on county exposure to large lenders confirm the **supply-side** interpretation.
- Among the mechanisms the authors consider, the explanation most consistent with all facts is that **post-crisis regulation increased fixed and per-loan origination costs**, which made small loans less attractive, especially for large lenders with a comparative advantage in jumbo lending.
- **Economic takeaway.** After the crisis, U.S. mortgage credit became more tilted toward larger loans and wealthier borrowers. The paper argues that this was not mainly a borrower-demand shift, but a **supply-driven reallocation of credit** amplified by large lenders.

## 2 Data and measurement (key objects)

### 2.1 HMDA data and baseline loan sample

- **Main data source.** The paper uses the **Home Mortgage Disclosure Act (HMDA)** data from the Consumer Financial Protection Bureau for **2007–2017**.
- **Coverage.** HMDA contains the vast majority of mortgage applications over the sample period.
- **Unit of observation.** An individual mortgage application.
- **Observed loan/applicant variables.**
  - requested loan amount,
  - approval/rejection status,
  - lien status,
  - loan purpose,
  - owner-occupancy status,
  - lender identifier,
  - applicant income,
  - applicant race and ethnicity,
  - county and census tract.
- **Main working sample.** The baseline individual-loan analysis keeps only:
  - **home-occupied new-purchase** mortgages,
  - secured by a **first lien**.
- **Rejected loans.** The preferred rejection definition uses only HMDA action code 3, that is, **denied applications**.

### 2.2 Additional data sources

- **Zillow.** County-level house prices and foreclosures.
- **HUD.** County-level median household income.

- **ACS/IPUMS.** Household income, labor-force participation, hours worked, homeownership, and rent for a representative 5% sample of the U.S. population.
- **BLS.** County unemployment rates.
- **BEA.** County GDP per full-time units.
- **U.S. Census Bureau.** County-level bank deposits.

These extra data sets are used both for demand controls and for the parallel-trends analysis behind the IV strategy.

## 2.3 Key definitions: lender size, loan-size bins, and conforming limits

- **Top-five lenders.** In a given year, these are the five largest U.S. lenders by overall mortgage origination activity.
- **County exposure to large lenders.** A key variable is the **share of county-level mortgage activity** accounted for by top-five lenders.
- **Loan-size bins in the raw-data analysis.**
  - below \$200K,
  - \$201K–\$400K,
  - \$401K–\$600K,
  - \$601K–\$800K,
  - above \$800K.
- **Baseline conforming loan limit.** The standard CLL was **\$417K** from 2008 through 2016 and rose to **\$424K** only in 2017.
- **High-cost exception counties.** The maximum conforming limit in these counties was **\$729.75K** in 2008–2010 and **\$625.5K** in 2011–2017.

## 2.4 Key descriptive facts from the data

- **Approved applications.** The sample contains **26,419,090** approved applications from 2007 to 2017.
- **Rejected applications.** It contains **4,954,288** rejected applications.
- **Approved-loan distribution.**
  - Median approved loan amount: **\$186K**.
  - 5th percentile: **\$64K**.
  - 95th percentile: **\$516K**.
- **Rejected-loan distribution.**
  - Median requested amount: **\$145K**.

- Rejected loans have lower median size than approved loans, but also fatter tails.
- **Top-five lenders.**
  - They account for about **14%** of approved applications.
  - They account for about **21%** of rejected applications.
- **Applicant observables.**
  - Median approved-applicant income: **\$71K**.
  - Mean approved-applicant income: **\$95.5K**.
  - Roughly **8%** of approved applicants are Black, **6%** Asian, and **10%** Latino.

## 2.5 Measurement issues and interpretation

- HMDA income is self-reported and may be noisy. The paper does *not* use income to estimate a structural borrower-demand model.
- Instead, income and demographic variables are used mainly as **controls for mortgage demand**.
- The distinctive strength of HMDA is that it includes both **approved and rejected** applications, which lets the authors study lenders' approval choices rather than just the equilibrium stock of originated loans.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 Extensive-margin approval model

The baseline approval specification is a linear probability model:

$$\text{Approved}_{i,b,t} = \alpha + \beta \log(\text{LoanSize}_{i,b,t}) + \gamma \log(\text{LoanSize}_{i,b,t}) \times \text{After2010}_t + X'_{i,b,t} \delta + \eta_t + \eta_k + \varepsilon_{i,b,t}.$$

- $\text{Approved}_{i,b,t}$ : indicator equal to one if application  $i$  to lender  $b$  in year  $t$  is approved.
- $\text{LoanSize}_{i,b,t}$ : requested dollar amount of the loan.
- $\text{After2010}_t$ : indicator for years **2011–2017**; the control period is **2008–2010**.
- $X_{i,b,t}$ : applicant and local controls.
- $\eta_t$ : year fixed effects.
- $\eta_k$ : county fixed effects, sometimes replaced by county-year fixed effects.

#### Interpretation.

- $\beta$  measures how approval varies with loan size in the pre-2011 period.
- $\gamma$  measures whether larger loans become more likely to be approved after 2010.

- This is a **difference-in-slopes** design: the authors are not asking whether overall approval rises or falls, but whether the approval schedule tilts toward larger loans after 2010.

### 3.2 Triple-interaction model for large lenders

The next specification lets the post-2010 tilt differ for top-five lenders:

$$\begin{aligned} \text{Approved}_{i,b,t} = & \alpha + \delta \log(\text{LoanSize}_{i,b,t}) \times \text{After2010}_t \times \text{Top5}_{b,t} \\ & + \beta \log(\text{LoanSize}_{i,b,t}) + \gamma \log(\text{LoanSize}_{i,b,t}) \times \text{After2010}_t \\ & + \omega \log(\text{LoanSize}_{i,b,t}) \times \text{Top5}_{b,t} + \zeta \text{After2010}_t \times \text{Top5}_{b,t} + \xi \text{Top5}_{b,t} \\ & + X'_{i,b,t} \theta + \eta_t + \eta_k + \varepsilon_{i,b,t}. \end{aligned}$$

- $\text{Top5}_{b,t}$ : indicator equal to one if lender  $b$  is one of the five largest U.S. originators in year  $t$ .
- The coefficient of interest is  $\delta$ .

#### Interpretation.

- $\delta$  is a **triple-difference-in-slopes** coefficient.
- It asks whether, after 2010, top-five lenders become more tilted toward larger loans *relative both to their own pre-2011 behavior and to smaller lenders' post-2010 behavior*.
- This is the paper's main reduced-form specification for the extensive margin.

### 3.3 Quantile regressions for the intensive margin

To study the dollar size of approved loans, the paper estimates quantile regressions of the form:

$$Q_\tau(\text{LoanSize}_{i,b,t}) = \alpha(\tau) + \beta(\tau) \text{After2010}_t + X'_{i,b,t} \zeta(\tau) + \varepsilon_{i,b,t},$$

estimated separately for top-five and non-top-five lenders.

#### Interpretation.

- $Q_\tau(\text{LoanSize})$  is the  $\tau$ -th quantile of the approved-loan-size distribution.
- $\beta(\tau)$  tells us by how much that quantile shifts after 2010.
- This is useful because the paper is about the **entire distribution** of loan sizes, not just the mean.
- If  $\beta(\tau)$  rises with  $\tau$ , the upper tail of the distribution is moving out more strongly than the middle or lower tail.

### 3.4 Cross-lender quantile comparison

To compare the distributional shift across large and small lenders directly, the paper estimates:

$$Q_{\tau}(\text{LoanSize}_{i,b,t}) = \alpha(\tau) + \beta(\tau) \text{Top5}_{b,t} \times \text{After2010}_t + \gamma(\tau) \text{Top5}_{b,t} + X'_{i,b,t} \zeta(\tau) + \eta_t + \varepsilon_{i,b,t}.$$

#### Interpretation.

- $\beta(\tau)$  is the differential post-2010 shift in the  $\tau$ -th quantile for top-five lenders relative to other lenders.
- Negative values below the conforming limit and positive values far above it mean that smaller lenders shift mostly within the conforming segment, whereas large lenders shift more strongly into jumbo lending.

### 3.5 Instrumental-variable specification

The extensive-margin IV specification is:

$$\begin{aligned} \text{Approved}_{i,b,t} = & \alpha + \beta \log(\text{LoanSize}_{i,b,t}) + \gamma \log(\text{LoanSize}_{i,b,t}) \times \text{After2010}_t \\ & + \delta \log(\text{LoanSize}_{i,b,t}) \times \text{After2010}_t \times \text{ShareTop5}_{c,t} \\ & + \zeta \text{After2010}_t \times \text{ShareTop5}_{c,t} + X'_{i,b,t} \theta + \eta_t + \eta_c + \varepsilon_{i,b,t}. \end{aligned}$$

- $\text{ShareTop5}_{c,t}$ : share of mortgage originations in county  $c$  and year  $t$  produced by top-five lenders.
- The endogenous county-year share is instrumented with the county's **share of top-five lenders in 2007**, before the crisis.

#### Interpretation.

- The IV design asks whether counties more exposed to large lenders for pre-crisis reasons experienced a stronger post-2010 tilt of approvals toward larger loans.
- The paper uses a **share** rather than a binary top-five indicator to avoid a forbidden-regression problem in 2SLS.

## 4 Identification and instruments (what makes the estimates credible)

### 4.1 What the raw patterns cannot identify

- Raw originations alone mix up borrower demand and lender supply.
- Large and small lenders do not serve the same borrowers in a random way.
- Changes around the conforming loan limit could, in principle, reflect bunching incentives rather than a deeper supply-side reallocation.

The paper's empirical strategy is designed precisely to separate these stories.



## 4.2 Why HMDA is central for identification

- HMDA includes both **approved and rejected** applications.
- This lets the authors condition on the overall pool of applicants and study lenders' **approval decisions** rather than only observed loan originations.
- By controlling for applicant income, race, local racial composition, local income, house prices, foreclosures, and fixed effects, the paper narrows the remaining variation to lender behavior.

## 4.3 Within-county and within-county-year comparisons

- County fixed effects absorb time-invariant local features.
- Year fixed effects absorb nationwide shocks.
- County-year fixed effects absorb local macro shocks and local credit-demand shifts.

This means identification often comes from comparing applications of different sizes **within the same local market and year**.

## 4.4 Instrument: pre-crisis exposure to large lenders

- The instrument is the **share of top-five lenders in a county in 2007**.
- **Relevance.** Bank branching and local lending structure are persistent because of relocation costs and geographic segmentation.
- **Exclusion restriction.** Conditional on controls, pre-crisis large-lender share should affect post-2010 mortgage approval patterns only through later exposure to large lenders, not through separate county shocks.

## 4.5 Evidence supporting the IV design

- Counties with higher and lower 2007 exposure to top-five lenders have very similar trends in:
  - household income,
  - GDP per capita,
  - unemployment,
  - labor-force participation,
  - hours worked,
  - rent,
  - deposit growth.
- The paper also reports balance tests showing observable determinants of mortgage demand are similar across counties with different values of the instrument.
- First-stage strength is high: the Kleibergen–Paap Wald  $F$ -statistics are about **45.2**, **17.6**, and **17.9** across the main specifications.

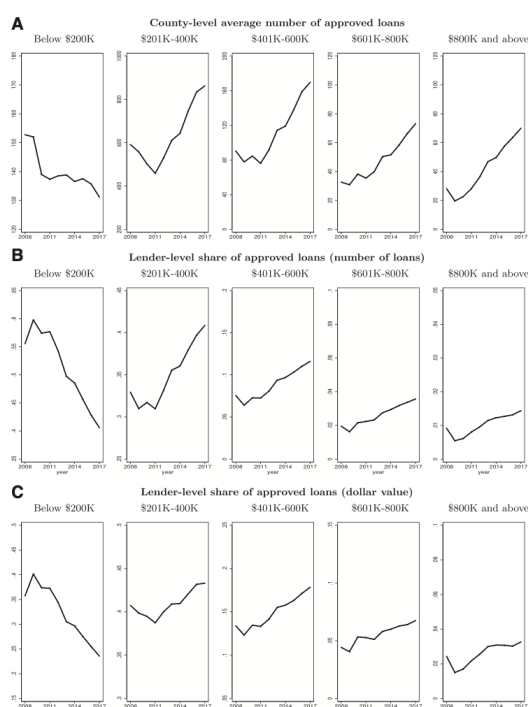
- The null of underidentification is rejected in all IV specifications.

## 4.6 What the IV results imply

- The IV estimates confirm the sign of the baseline OLS results.
- When the paper adds county-year fixed effects, the IV coefficients become larger in magnitude, which the authors interpret as consistent with a stronger causal supply-side effect once more local confounding variation is absorbed.

## 5 Tables and figures

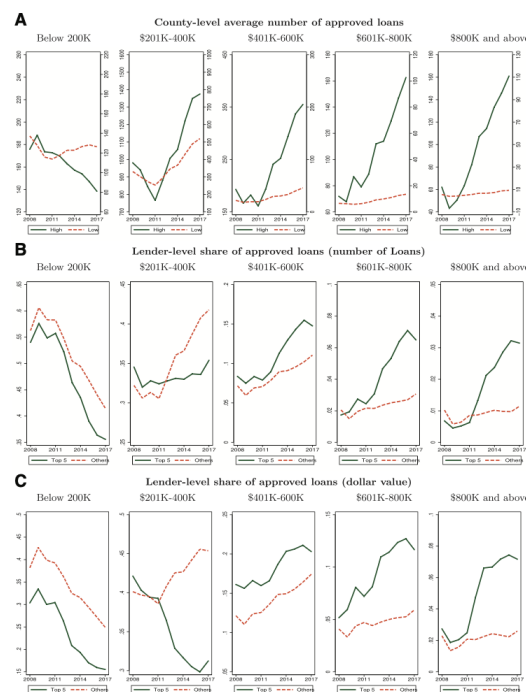
### 5.1 Figure 1



**Figure 1**  
This figure reports the change in lending by loan size bins for the years 2008-2017. Panel A reports the average number of approved loans, computed at the county level. Panels B and C reports the share of approved loans, computed at the lender level. In panel B the share is computed based on the overall number of approved loans. In panel C, it is computed based on the overall dollar value of approved loans. Within each panel, we report five plots for the following size bins: below \$200K, \$201K-\$400K, \$401K-\$600K, \$601K-\$800K, and \$800K and above.

### Read:

- The number of mortgages below \$200K falls over time, especially after 2011.
- Larger size bins rise steadily after 2011.
- Within lenders, the share of loans below \$200K drops from about **57%** in 2011 to **41%** in 2017.
- The share of loans between \$201K and \$400K rises from about **31%** to **43%**.



**Figure 2**  
This figure reports the change in lending by loan size bins for the years 2008-2017, separately by lender size. Panel A reports the average number of approved loans, computed at the county level, across counties with a share of top-five lenders in the top-two quintiles and other counties. Panels B and C reports the share of approved loans, computed at the lender level, across top-five lenders and other lenders. In panel B the share is computed based on the overall number of approved loans. In panel C, it is computed based on the overall dollar value of approved loans. Within each panel, we report five plots for the following size bins: below \$200K, \$201K-\$400K, \$401K-\$600K, \$601K-\$800K, and \$800K and above.

- The share of the largest loans (above \$800K) rises from about **0.8%** to **1.4%**.

**Punchline.** The aggregate U.S. mortgage distribution shifts to the right after 2011.

## 5.2 Figure 2

**Read:**

- Counties with greater exposure to top-five lenders drive more of the drop in small loans.
- Top-five lenders increase their origination of large loans much more than smaller lenders do.
- Smaller lenders mostly move toward somewhat larger loans *within* the conforming segment.

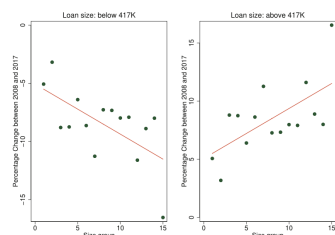
**Punchline.** The mortgage-credit redistribution is much stronger among large lenders.

## 5.3 Figure 3

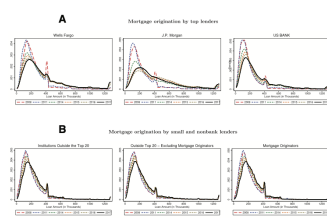
**Read:**

- For loans below \$417K, larger lenders show a stronger decline in origination from 2008 to 2017.
- For loans above \$417K, larger lenders show a stronger increase in origination.
- The relationship is smooth and monotonic across lender-size groups.

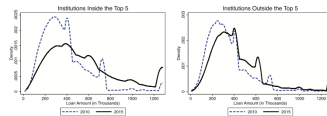
**Punchline.** The effect is proportional to lender size, not driven by one special size threshold.



**Figure 3**  
This figure plots the raw percentage change in loans originated by lender size between 2008 and 2017. The left panel considers loans below \$417K. The right panel considers loans above \$417K. We group lenders in 15 equal size groups based on total lending, and report the value-weighted change in lending for each group. We limit the sample to the 1,000 top lenders.



**Figure 4**  
Panel A reports the loan size distribution by three top originators by lending activity over the sample. Wells Fargo, JP Morgan, and U.S. Bank. Panel B reports the loan size distributions for all financial institutions that rank outside the top 20 (left plot), for the bank originators that rank outside the top 20 (middle plot), and for nonbank mortgage originators (right plot). Each plot reports densities for the years 2008, 2011, 2013, 2015, 2014, 2015, 2016, and 2017. Loan amounts are winsorized at the 0.5% level.



**Figure 5**  
The left plot shows the loan size distribution in counties with exceptions to the \$417K conforming loan limit in 2010 and 2015 for loans originated by the top 5 lenders based on U.S.-wide total origination. The right plot shows the loan size distribution in counties with exceptions to the \$417K conforming loan limit in 2010 and 2015 for loans originated by all institutions outside the top-20 lenders based on U.S.-wide total origination. Loan amounts are winsorized at the 0.5% level.

## 5.4 Figures 4 and 5

**Read:**

- In 2008, large lenders show strong bunching just below the conforming loan limit.
- After 2011, large lenders' distributions move rightward and bunching near the conforming limit shrinks sharply or disappears.
- Smaller lenders and nonbank originators still bunch around conforming limits and mainly move within the conforming range.
- The same logic holds in high-cost counties with higher conforming limits.

**Punchline.** The results are not simply a story about bunching at the conforming limit; large lenders move beyond the conforming segment altogether.

## 5.5 Table 2

**Read:**

- In the baseline approval regression, the coefficient on log loan size is about **0.038**, and the post-2010 slope change is about **0.021**.
- With richer controls and county-year fixed effects, the post-2010 slope change remains positive at about **0.014**.
- In the triple-interaction specification, the coefficient on  $\log(\text{loan size}) \times \text{After2010} \times \text{Top5}$  is around **0.014** and highly significant.

**Punchline.** Even conditional on observables and local macro shocks, larger loans are more likely to be approved after 2010, especially by top-five lenders.

**Table 2**  
Mortgage origination in the post-crisis period: Extensive margin (loan approval)

|                                                                          | (1)                | (2)                | (3)                | (4)                | (5)                | (6)                |
|--------------------------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| log(loan) size ( $\beta$ )                                               | 0.038<br>(11.59)   | 0.044<br>(14.27)   | 0.044<br>(14.28)   | 0.043<br>(12.43)   | 0.048<br>(15.20)   | 0.048<br>(15.22)   |
| log(loan) size $\times$<br>After 2010 ( $\gamma$ )                       | 0.021<br>(12.19)   | 0.014<br>(9.91)    | 0.014<br>(9.74)    | 0.02<br>(10.39)    | 0.012<br>(7.93)    | 0.012<br>(7.76)    |
| log(loan) size $\times$<br>After 2010 $\times$ Top 5 lender ( $\delta$ ) |                    |                    |                    | 0.014<br>(5.29)    | 0.013<br>(5.44)    | 0.014<br>(5.58)    |
| log(loan) size $\times$<br>Top 5 lender ( $\omega$ )                     |                    |                    |                    | -0.035<br>(-11.62) | -0.034<br>(-11.63) | -0.035<br>(-11.69) |
| After 2010 $\times$<br>Top 5 lender ( $\zeta$ )                          |                    |                    |                    | -0.116<br>(-9.32)  | -0.121<br>(-10.03) | -0.122<br>(-10.19) |
| Top 5 lender ( $\xi$ )                                                   |                    |                    |                    | 0.17<br>(10.72)    | 0.168<br>(10.99)   | 0.168<br>(11.05)   |
| log(income)                                                              | 0.042<br>(39.70)   | 0.042<br>(40.48)   | 0.043<br>(40.44)   | 0.044<br>(41.52)   | 0.044<br>(42.24)   | 0.045<br>(42.26)   |
| Black                                                                    | -0.104<br>(-53.80) | -0.104<br>(-54.05) | -0.104<br>(-54.64) | -0.104<br>(-54.39) | -0.103<br>(-54.87) | -0.104<br>(-55.46) |
| Asian                                                                    | -0.030<br>(-12.03) | -0.029<br>(-12.03) | -0.030<br>(-12.09) | -0.027<br>(-10.86) | -0.027<br>(-10.84) | -0.027<br>(-10.89) |
| Latino                                                                   | -0.037<br>(-20.89) | -0.037<br>(-20.63) | -0.037<br>(-20.89) | -0.038<br>(-22.33) | -0.038<br>(-22.06) | -0.038<br>(-22.31) |
| Share minority                                                           | 0.000              | -0.000             | -0.000             | 0.000              | 0.000              | -0.000             |
| Population (census tract)                                                | (-2.65)            | (-2.66)            | (-2.27)            | (-3.27)            | (-3.33)            | (-2.92)            |
| Log(median<br>house price county)                                        | 0.014<br>(2.06)    |                    |                    | 0.017<br>(2.31)    |                    |                    |
| Male                                                                     |                    |                    | -0.002<br>(-3.81)  |                    |                    | -0.002<br>(-4.03)  |
| Number of 1-4 family<br>Unit (ZIP code)                                  |                    |                    | 0.000<br>(8.72)    |                    |                    | 0.000<br>(8.92)    |
| Constant                                                                 | 0.378<br>(10.49)   | 0.407<br>(26.64)   | 0.400<br>(26.50)   | 0.340<br>(9.41)    | 0.391<br>(24.91)   | 0.384<br>(24.74)   |
| County FE                                                                | X                  |                    |                    | X                  |                    |                    |
| Year FE                                                                  | X                  |                    |                    | X                  |                    |                    |
| County $\times$ Year FE                                                  |                    | X                  | X                  |                    | X                  | X                  |
| Observations                                                             | 27,706,874         | 27,706,874         | 27,633,649         | 27,693,907         | 27,693,907         | 27,620,737         |
| Adj. $R^2$                                                               | 0.059              | 0.062              | 0.062              | 0.062              | 0.065              | 0.065              |

This table reports results on the extensive margin of mortgage origination: the decision to approve or reject a loan application. In columns 1–3, the analysis is based on estimating linear probability models of the following form:

$$\text{Approved}_{i,b,t} = \alpha + \beta \text{Loan size}_{i,b,t} + \gamma \text{Loan size}_{i,b,t} \times \text{After 2010}_t + X'_{i,b,t} \delta + \eta_t + \eta_k + \epsilon_{i,b,t},$$

**Table 3**  
Mortgage origination in the post-crisis period: Extensive margin (loan approval): Robustness

|                                                                          | (1)<br>Full interactions<br>controls | (2)<br>Concentration<br>lending | (3)<br>Foreclosures<br>ZIP code | (4)<br>Excluding<br>mortg. originators | (5)<br>Excluding<br>sand states |
|--------------------------------------------------------------------------|--------------------------------------|---------------------------------|---------------------------------|----------------------------------------|---------------------------------|
| log(loan size) $\times$<br>After 2010 $\times$ Top 5 lender ( $\delta$ ) | 0.016<br>(6.88)                      | 0.014<br>(5.58)                 | 0.019<br>(6.79)                 | 0.023<br>(9.59)                        | 0.016<br>(6.67)                 |
| Other interactions                                                       | X                                    | X                               | X                               | X                                      | X                               |
| Baseline controls                                                        | X                                    | X                               | X                               | X                                      | X                               |
| Additional controls                                                      | X                                    | X                               | X                               | X                                      | X                               |
| County $\times$ Year FE                                                  | X                                    | X                               | X                               | X                                      | X                               |
| Observations                                                             | 27,620,737                           | 27,620,737                      | 16,871,168                      | 13,281,075                             | 21,743,435                      |
| Adj. $R^2$                                                               | 0.065                                | 0.065                           | 0.050                           | 0.060                                  | 0.073                           |

This table reports results on the extensive margin of mortgage origination: the decision to approve or reject a loan application. The analysis is based on estimating linear probability models of the following form:

$$\begin{aligned} \text{Approved}_{i,b,t} = & \alpha + \beta \log(\text{loan size})_{i,b,t} + \gamma \log(\text{loan size})_{i,b,t} \times \text{After 2010}_t \\ & + \delta \log(\text{loan size})_{i,b,t} \times \text{After 2010}_t \times \text{Top 5 lender}_t \\ & + \zeta \text{After 2010}_t \times \text{Top 5 lender}_t + \xi \text{Top 5 lender}_t \\ & + X'_{i,b,t} \theta + \eta_t + \epsilon_{i,b,t}, \end{aligned}$$

where  $\text{Approved}_{i,b,t}$  is a dummy variable that equals one if loan application  $i$  to bank  $b$  in year  $t$  was approved, and zero otherwise;  $\log(\text{loan size})_{i,b,t}$  is the logarithm of the dollar amount requested in loan application  $i$ ;  $\text{After 2010}_t$  is a dummy variable that equals one in years 2011–2017, and zero in years 2008–2010; and  $\text{Top 5 lender}_t$  is a dummy variable that equals one for lenders that are among the top-five in the United States by overall origination activity in year  $t$ , and zero otherwise.  $X'_{i,b,t}$  is a set of borrower characteristics we observe in HMDA;  $\eta_t$  is a full set of county-year fixed effects (note the year fixed effects absorb the level of the variable  $\text{After 2010}_t$ ).

## 5.6 Table 3

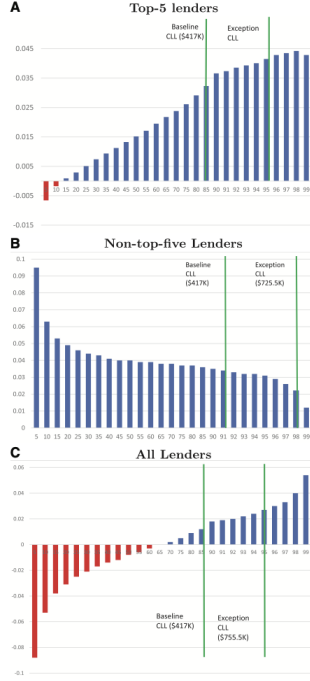
**Read:**

- The main triple interaction remains robust when the paper:
  - allows fully interacted controls,
  - controls for local lending concentration,
  - controls for foreclosures,

- excludes nonbank mortgage originators,
- excludes sand states.

**Punchline.** The baseline extensive-margin result is not an artifact of a narrow specification.

## 5.7 Figure 6 and Table 4



**Figure 6**  
This figure plots the coefficients  $\hat{\beta}(\tau)$  estimated from a set of quantile regressions of the following form:

$$Q_{\tau}(Loan\ size_{b,t}) = \alpha(\tau) + \beta(\tau) After\ 2010_t \times Top\ 5\ lender_{b,t} + \gamma(\tau) Top\ 5\ lender_{b,t} + X'_{i,b,t} \zeta(\tau) + \eta_{kt} + \epsilon_{b,t},$$

where  $Q_{\tau}(Loan\ size_{b,t})$  is the  $g$ th quantile of the distribution of the size of approved loans by lender  $b$  in year  $t$ ;  $Top\ 5\ lender_{b,t}$  is a dummy variable that equals one if lender  $b$  in year  $t$  is one of the top-five U.S. originators nationally, and zero otherwise;  $After\ 2010_t$  is a dummy variable that equals one in years 2011–2017, and zero in years 2008–2010; and  $X'_{i,b,t}$  is a set of average borrower characteristics at the lender and year level we observe in HMDA;  $\eta_{kt}$  is a full set of county-year fixed effects (note the year fixed effects absorb the level of the variable  $After\ 2010_t$ ). In panel A, we estimate these specifications only for the loans originated by the top-five lenders based on overall U.S. mortgage origination. In panel B, we restrict the sample to the loans originated by lenders outside the top-five group. In both of these panels, we plot the coefficient attached to the  $After\ 2010_t$  dummy. In panel C, we estimate these specifications for all loans in the sample, and plot the coefficient attached to the interaction  $After\ 2010_t \times Top\ 5\ lender_{b,t}$ .

**Read:**

- For top-five lenders, nearly the whole loan-size distribution shifts right after 2010, and the shift becomes larger at higher quantiles.
- For non-top-five lenders, the right shift is still present, but it is stronger at lower and middle quantiles and weaker in the upper tail.
- In the direct comparison, the differential top-five effect is negative below the conforming range and positive in the upper tail.

**Punchline.** Large lenders disproportionately expand in very large loans, while smaller lenders shift mainly within conforming loans.

**Table 4**  
Mortgage origination in the post-crisis period: Intensive margin (distribution loan sizes)

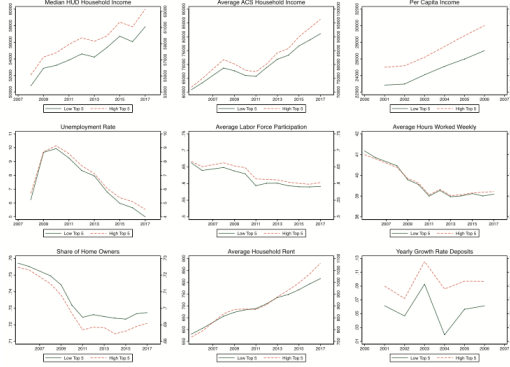
| Percentile          | Top 5 lenders |                    |        | Non-Top 5 lenders |                    |       | All lenders |                    |        |
|---------------------|---------------|--------------------|--------|-------------------|--------------------|-------|-------------|--------------------|--------|
|                     | Coefficient   | 95% conf. interval |        | Coefficient       | 95% conf. interval |       | Coefficient | 95% conf. interval |        |
| 5th                 | -0.007        | -0.010             | -0.003 | 0.095             | 0.093              | 0.098 | -0.088      | -0.092             | -0.084 |
| 10th                | -0.002        | -0.004             | 0.001  | 0.064             | 0.063              | 0.066 | -0.053      | -0.056             | -0.050 |
| 15th                | 0.001         | -0.001             | 0.003  | 0.054             | 0.053              | 0.055 | -0.038      | -0.040             | -0.036 |
| 20th                | 0.003         | 0.001              | 0.005  | 0.050             | 0.049              | 0.051 | -0.031      | -0.033             | -0.029 |
| 25th                | 0.005         | 0.004              | 0.007  | 0.047             | 0.046              | 0.048 | -0.025      | -0.027             | -0.024 |
| 30th                | 0.007         | 0.006              | 0.009  | 0.045             | 0.044              | 0.045 | -0.021      | -0.022             | -0.019 |
| 35th                | 0.009         | 0.008              | 0.011  | 0.043             | 0.043              | 0.044 | -0.017      | -0.019             | -0.016 |
| 40th                | 0.011         | 0.010              | 0.012  | 0.042             | 0.041              | 0.043 | -0.014      | -0.016             | -0.013 |
| 45th                | 0.013         | 0.012              | 0.014  | 0.041             | 0.040              | 0.044 | -0.012      | -0.013             | -0.010 |
| 50th                | 0.015         | 0.014              | 0.016  | 0.040             | 0.040              | 0.041 | -0.008      | -0.010             | -0.007 |
| 55th                | 0.017         | 0.016              | 0.018  | 0.040             | 0.039              | 0.040 | -0.006      | -0.007             | -0.004 |
| 60th                | 0.020         | 0.018              | 0.021  | 0.039             | 0.039              | 0.040 | -0.003      | -0.004             | -0.002 |
| 65th                | 0.022         | 0.021              | 0.023  | 0.038             | 0.038              | 0.039 | -0.000      | -0.002             | 0.000  |
| 70th                | 0.024         | 0.023              | 0.025  | 0.038             | 0.037              | 0.039 | 0.002       | 0.001              | 0.003  |
| 75th                | 0.026         | 0.025              | 0.027  | 0.037             | 0.037              | 0.038 | 0.005       | 0.004              | 0.006  |
| 80th                | 0.029         | 0.028              | 0.030  | 0.037             | 0.036              | 0.038 | 0.009       | 0.007              | 0.010  |
| 85th                | 0.032         | 0.031              | 0.033  | 0.036             | 0.035              | 0.037 | 0.012       | 0.011              | 0.014  |
| 90th                | 0.037         | 0.035              | 0.038  | 0.035             | 0.034              | 0.035 | 0.018       | 0.016              | 0.019  |
| 91st                | 0.037         | 0.036              | 0.039  | 0.034             | 0.033              | 0.035 | 0.019       | 0.018              | 0.020  |
| 92nd                | 0.039         | 0.037              | 0.040  | 0.033             | 0.033              | 0.034 | 0.020       | 0.019              | 0.022  |
| 93rd                | 0.039         | 0.038              | 0.041  | 0.032             | 0.032              | 0.033 | 0.022       | 0.021              | 0.024  |
| 94th                | 0.040         | 0.039              | 0.041  | 0.032             | 0.031              | 0.033 | 0.024       | 0.023              | 0.026  |
| 95th                | 0.042         | 0.040              | 0.043  | 0.031             | 0.030              | 0.031 | 0.027       | 0.025              | 0.028  |
| 96th                | 0.043         | 0.041              | 0.044  | 0.029             | 0.028              | 0.029 | 0.030       | 0.028              | 0.032  |
| 97th                | 0.044         | 0.042              | 0.045  | 0.026             | 0.025              | 0.027 | 0.033       | 0.031              | 0.035  |
| 98th                | 0.044         | 0.042              | 0.046  | 0.022             | 0.021              | 0.023 | 0.040       | 0.038              | 0.042  |
| 99th                | 0.043         | 0.040              | 0.045  | 0.012             | 0.010              | 0.013 | 0.054       | 0.051              | 0.057  |
| Baseline controls   | X             | X                  | X      | X                 | X                  | X     | X           | X                  | X      |
| Additional controls | X             | X                  | X      | X                 | X                  | X     | X           | X                  | X      |
| County × Year FE    | X             | X                  | X      | X                 | X                  | X     | X           | X                  | X      |
| Observations        | 2,293,268     |                    |        | 10,914,549        |                    |       | 13,208,431  |                    |        |

This table reports results on the intensive margin of mortgage origination: the distribution of the size of approved loan applications over time. The analysis is based on estimating the following set of quantile regressions of the following form:

$$Q_{\tau}(Loan\ size_{b,t}) = \alpha(\tau) + \beta(\tau) After\ 2010_t \times Top\ 5\ lender_{b,t} + \gamma(\tau) Top\ 5\ lender_{b,t} + X'_{i,b,t} \zeta(\tau) + \eta_{kt} + \epsilon_{b,t},$$

where  $Q_{\tau}(Loan\ size_{b,t})$  is the  $g$ th quantile of the distribution of the size of approved loans by bank  $b$  in year  $t$ ;  $Top\ 5\ lender_{b,t}$  is a dummy variable that equals one if bank  $b$  in year  $t$  is one of the top-five U.S. originators nationally, and zero otherwise;  $After\ 2010_t$  is a dummy variable that equals one in years 2011–2017, and zero in years 2008–2010;  $X'_{i,b,t}$  is a set of average borrower characteristics at the bank and year level we observe in HMDA; and  $\eta_{kt}$  is a full set of county-year fixed effects (note the year fixed effects absorb the level of the variable  $After\ 2010_t$ ).

## 5.8 Figure 9 and Table 5



**Figure 9**  
This figure plots the average of nine county-level economic variables from 2008 to 2017 across two groups of counties: counties in the top-two quintiles by the share of the top-five lenders in the county (High Top 5, right y-axis) and other counties (Low Top 5, left y-axis). In every plot, the two axes have the same range to ensure the trends can be compared directly. We describe the source of each variable in the main text.

**Table 5**  
Instrumental variable strategy: Extensive margin (loan approval)

|                                                     | Second-stage results |            |            |
|-----------------------------------------------------|----------------------|------------|------------|
|                                                     | (1)                  | (2)        | (3)        |
| $\log(\text{loan size}) \times$                     | 0.018                | 0.155      | 0.153      |
| After 2010 $\times$ Share Top 5 lender ( $\delta$ ) | (2.44)               | (3.63)     | (3.66)     |
| Other (instrumented) interactions                   | X                    | X          | X          |
| Baseline controls                                   | X                    | X          | X          |
| Additional controls                                 |                      |            | X          |
| County FE                                           | X                    |            |            |
| Year FE                                             | X                    |            |            |
| County $\times$ Year FE                             |                      | X          | X          |
| Observations                                        | 27,693,912           | 27,693,686 | 27,620,516 |
| First-stage hypothesis testing                      |                      |            |            |
|                                                     | (1)                  | (2)        | (3)        |
| Kleibergen-Paap Wald F-statistic                    | 45.20                | 17.63      | 17.87      |
| Kleibergen-Paap LM-test statistic                   | 56.12                | 20.13      | 20.31      |
| <i>p</i> -value                                     | (0.00)               | (0.00)     | (0.00)     |

This table reports the first- and second-stage results on the extensive margin of mortgage origination—the decision to approve or reject a loan application. We estimate linear probability models of the following form:

$$\begin{aligned}
 \text{Approved}_{i,b,t} = & \alpha + \beta \log(\text{loan size})_{i,b,t} + \gamma \log(\text{loan size})_{i,b,t} \times \text{After } 2010_t \\
 & + \delta \log(\text{loan size})_{i,b,t} \times \widehat{\text{Share top 5 lender}}_{i,b,t} \\
 & + \zeta \text{After } 2010_t \times \widehat{\text{Share Top 5 lender}}_{i,b,t} \\
 & + X'_{i,b,t} \theta + \eta_t + \eta_k + \epsilon_{i,b,t},
 \end{aligned}$$

where  $\text{Approved}_{i,b,t}$  is a dummy variable that equals one if loan application  $i$  to bank  $b$  in year  $t$  was approved, and zero otherwise;  $\log(\text{loan size})_{i,b,t}$  is the logarithm of the dollar amount requested in loan application  $i$ ;  $\text{After } 2010_t$  is a dummy variable that equals one in years 2011–2017, and zero in years 2008–2010; and  $\widehat{\text{Share Top 5 lender}}_{i,b,t}$  is the share of county-level originations produced by top-five lenders in the county and year in which the loan is originated. We instrument all the interaction terms that include the term  $\widehat{\text{Share Top 5 lender}}_{i,b,t}$  with the related interaction that substitutes  $\widehat{\text{Share Top 5 lender}}_{i,b,t}$  with the share of top-five lenders in each U.S. county in 2007, before the financial crisis. For each interaction, we have a first stage regression;  $X'_{i,b,t}$  is a set of borrower characteristics we observe in HMDA, which we allow to vary systematically from 2011 onward, relative to before 2011;  $\eta_t$  is a full set of year fixed effects (note the year fixed effects absorb the level of the variable  $\text{After } 2010_t$ ); and  $\eta_k$  represents county fixed effects. The top panel of the table reports the second-stage coefficient estimates of interest ( $\delta$ ). The bottom panel of the table reports, for each specification, the Kleibergen-Paap Wald F-statistic to test for the weakness of the instrument and the Kleibergen-Paap statistic for the LM test for underidentification and its associated  $p$ -value.

**Read:**

- Counties with high and low pre-crisis exposure to top-five lenders display very similar trends in income, GDP, labor-market conditions, rent, and deposits.
- The IV second stage yields a positive and significant coefficient on the post-2010 exposure of large lenders.
- The IV coefficients are similar in sign and generally larger than OLS coefficients.

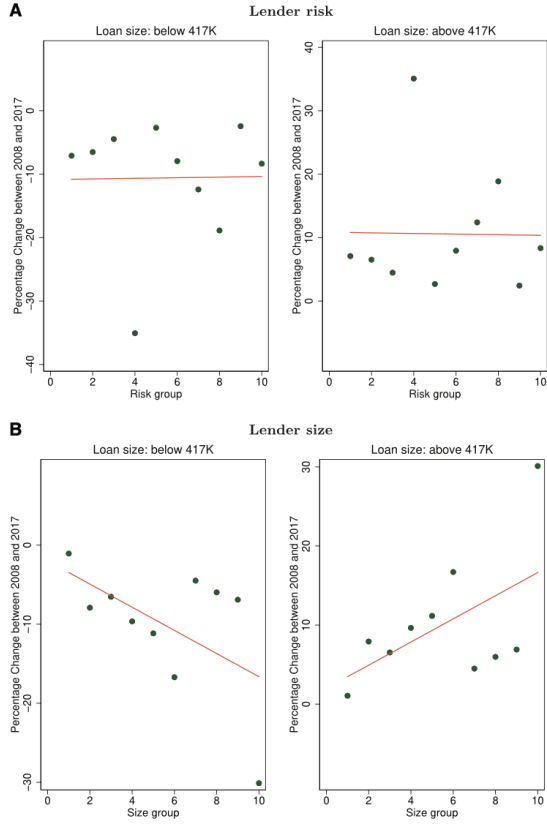
**Punchline.** The evidence supports a **supply-side** interpretation rather than a pure demand-composition story.

## 5.9 Figure 10 and Table 6

**Read:**

- Re-sorting lenders by bank risk rather than size does not reproduce the main result.
- The regressive redistribution is not stronger in years when QE-related MBS purchases were most effective.

**Punchline.** Simple risk-management and QE-channel explanations do not match the data well.



**Figure 10**  
This figure reports in panel A the raw percentage change in loans originated by lender risk between 2008 and 2017. The left panel considers loans below \$417K. The right panel considers loans above \$417K. The measure of lender risk is the share of reserves over the total amount of nonperforming loans held by the bank. Because we do not observe this measure of lender risk for all the institutions in our sample, we focus on the 100 riskiest institutions and group them into 10 equally sized groups based on lender risk. We report the value-weighted change in lending for each group. Panel B uses the same sample of lenders, but is sorted on size.

**Table 6**  
Change in mortgage origination by waves of quantitative easing: MBS purchases by the Fed

|                                  | Years QE most effective |            |            | Years QE least effective |            |            |
|----------------------------------|-------------------------|------------|------------|--------------------------|------------|------------|
|                                  | (1)                     | (2)        | (3)        | (4)                      | (5)        | (6)        |
| $\log(\text{loan size}) \times$  | 0.010                   | 0.011      | 0.011      | 0.016                    | 0.015      | 0.016      |
| After 2010 $\times$ Top 5 lender | (4.02)                  | (4.57)     | (4.69)     | (5.58)                   | (5.76)     | (5.89)     |
| Baseline controls                | X                       | X          | X          | X                        | X          | X          |
| Additional controls              |                         |            | X          |                          |            | X          |
| County FE                        | X                       |            |            | X                        |            |            |
| Year FE                          | X                       |            |            | X                        |            |            |
| County $\times$ Year FE          |                         | X          | X          |                          | X          | X          |
| Observations                     | 15,036,203              | 15,036,203 | 14,995,856 | 19,996,864               | 19,996,864 | 19,944,101 |
| $R^2$                            | 0.0576                  | 0.0611     | 0.0611     | 0.0593                   | 0.0630     | 0.0628     |

This table reports results on the extensive margin of mortgage origination: the decision to approve or reject a loan application. The analysis is based on estimating linear probability models of the following form:

$$\begin{aligned}
 \text{Approved}_{i,b,t} = & \alpha + \beta \log(\text{loan size})_{i,b,t} + \gamma \log(\text{loan size})_{i,b,t} \times \text{After } 2010_t \\
 & + \delta \log(\text{loan size})_{i,b,t} \times \text{After } 2010_t \times \text{Top } 5 \text{ lender}_{b,t} \\
 & + \zeta \text{After } 2010_t \times \text{Top } 5 \text{ lender}_{b,t} + \xi \text{Top } 5 \text{ lender}_{b,t} \\
 & + X'_{i,b,t} \theta + \eta_{kt} + \epsilon_{i,b,t},
 \end{aligned}$$

where  $\text{Approved}_{i,b,t}$  is a dummy variable that equals one if loan application  $i$  to bank  $b$  in year  $t$  was approved, and zero otherwise;  $\log(\text{loan size})_{i,b,t}$  is the logarithm of the dollar amount requested in loan application  $i$ ;  $\text{After } 2010_t$  is a dummy variable that equals one in years 2011-2017, and zero in years 2008-2010;  $\text{Top } 5 \text{ lender}_{b,t}$  is a dummy variable that equals one for lenders that are among the top five in the United States by overall origination activity in year  $t$ , and zero otherwise.  $X'_{i,b,t}$  is a set of borrower characteristics we observe in HMDA; and  $\eta_{kt}$  is a full set of county-year fixed effects (note the year fixed effects absorb the level of the variable  $\text{After } 2010_t$ ). The first three columns estimate the results using the post-2010 years in which QE was most effective (i.e., 2011, 2013, and 2014). The last three columns estimate the results using the post-2010 years in which QE was least effective (i.e., 2012, 2015, 2016, and 2017). We follow Chakraborty, Goldstein, and MacKinlay (2020) to categorize as effective those years that follow an MBS purchase wave in the third quarter of the previous year, and ineffective the other years. Note the for each group 2008, 2009, and 2010 are the control period based on our conjectures. Because 2 of these 3 years also had effective MBS purchase rounds, and Chakraborty, Goldstein, and MacKinlay (2020) find QE waves had stronger effects in the first part of the crisis relative to the second part, if anything our results estimate a lower bound of the effect of lender size on the likelihood of approval.

## 6 Short summary

- **Method.** The paper combines raw distributional evidence, approval-rate regressions, quantile regressions, and an IV design based on pre-crisis county exposure to large lenders.
- **Identification.** Rejected applications from HMDA help control for mortgage demand, while the 2007 county share of large lenders instruments for later exposure to top-five lenders.
- **Application.** U.S. mortgage origination in the post-crisis era, with special emphasis on how lending differs by loan size and lender size.
- **Main result.** Since 2011, U.S. mortgage credit has been reallocated away from smaller loans and toward larger loans, and this reallocation is strongest for the largest lenders and is best explained by a supply-side shift in lender incentives.

## 7 Conclusion

### 7.1 Core conclusion (what the paper establishes)

- The paper documents a persistent **regressive redistribution of mortgage credit** in the United States after the financial crisis.
- The change is not just that average loan size rises. Instead, the entire approval and origination schedule tilts away from small and midsize loans and toward large loans.
- The evidence is most consistent with a **supply-driven** shift, especially among the largest lenders.

### 7.2 Economic intuition

- If the fixed and per-loan costs of mortgage origination rise, lenders have an incentive to favor larger loans because the same compliance and underwriting costs are spread over a larger principal amount.
- Large lenders are especially well positioned to exploit this margin because they have larger balance sheets, broader product bundles, and a comparative advantage in serving jumbo borrowers.

### 7.3 Takeaway

The paper's main lesson is that post-crisis mortgage-market reforms may have had an unintended distributional side effect: they improved safety, but also pushed credit away from smaller loans and toward larger mortgages. In that sense, the mortgage market became more selective in a way that disadvantaged households needing modest-size loans, especially in areas dominated by large lenders.